

ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras

Raúl Mur-Artal and Juan D. Tardós

Abstract—We present ORB-SLAM2 a complete SLAM system for monocular, stereo and RGB-D cameras, including map reuse, loop closing and relocalization capabilities. The system works in real-time on standard CPUs in a wide variety of environments from small hand-held indoors sequences, to drones flying in industrial environments and cars driving around a city. Our back-end based on bundle adjustment with monocular and stereo observations allows for accurate trajectory estimation with metric scale. Our system includes a lightweight localization mode that leverages visual odometry tracks for unmapped regions and matches to map points that allow for zero-drift localization. The evaluation on 29 popular public sequences shows that our method achieves state-of-the-art accuracy, being in most cases the most accurate SLAM solution. We publish the source code, not only for the benefit of the SLAM community, but with the aim of being an out-of-the-box SLAM solution for researchers in other fields.

I. INTRODUCTION

Simultaneous Localization and Mapping (SLAM) has been a hot research topic in the last two decades in the Computer Vision and Robotics communities, and has recently attracted the attention of high-technological companies. SLAM techniques build a map of an unknown environment and localize the sensor in the map with a strong focus on real-time operation. Among the different sensor modalities, cameras are cheap and provide rich information of the environment that allows for robust and accurate place recognition. Therefore Visual SLAM solutions, where the main sensor is a camera, are of major interest nowadays. Place recognition is a key module of a SLAM system to close loops (i.e. detect when the sensor returns to a mapped area and correct the accumulated error in exploration) and to relocalize the camera after a tracking failure, due to occlusion or aggressive motion, or at system re-initialization.

Visual SLAM can be performed by using just a monocular camera, which is the cheapest and smallest sensor setup. However as depth is not observable from just one camera, the scale of the map and estimated trajectory is unknown. In addition the system bootstrapping require multi-view or filtering techniques to produce an initial map as it cannot be triangulated from the very first frame. Last but not least,

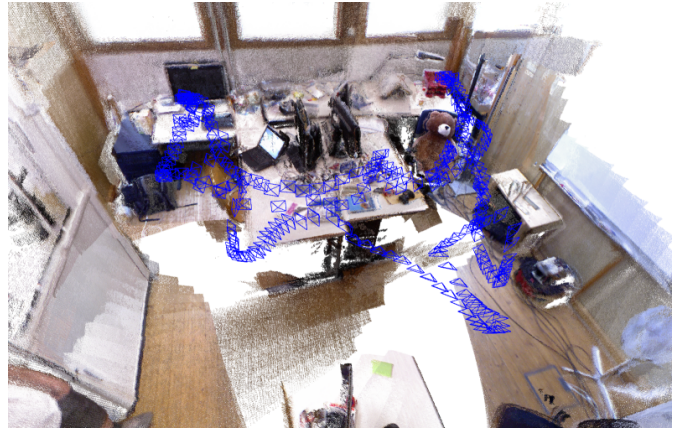
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(a) Stereo input: trajectory and sparse reconstruction of an urban environment with multiple loop closures.



(b) RGB-D input: keyframes and dense pointcloud of a room scene with one loop closure. The pointcloud is rendered by backprojecting the sensor depth maps from estimated keyframe poses. No fusion is performed.

Fig. 1. ORB-SLAM2 processes stereo and RGB-D inputs to estimate camera trajectory and build a map of the environment. The system is able to close loops, relocalize, and reuse its map in real-time on standard CPUs with high accuracy and robustness.

monocular SLAM suffers from scale drift and may fail if performing pure rotations in exploration. By using a stereo or an RGB-D camera all these issues are solved and allows for the most reliable Visual SLAM solutions.

In this paper we build on our monocular ORB-SLAM [1] and propose ORB-SLAM2 with the following contributions:

ORB-SLAM2：用于单目、立体和 RGB-D 相机的开源 SLAM 系统

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Abstract—我们推出了 ORB-SLAM2，这是一个适用于单目、立体和 RGB-D 相机的完整 SLAM 系统，包括地图重用、闭环和重新定位功能。该系统在各种环境下在标准 CPU 上实时工作，从小型手持式室内序列到在工业环境中飞行的无人机和在城市中行驶的汽车。我们的后端基于单目和立体观测的捆绑调整，可以使用公制比例进行准确的轨迹估计。我们的系统包括一个轻量级定位模式，该模式利用视觉里程计跟踪未映射的区域并匹配允许零漂移定位的地图点。对 29 个流行公共序列的评估表明，我们的方法达到了最先进的精度，在大多数情况下是最准确的 SLAM 解决方案。我们发布源代码，不仅是为了 SLAM 社区的利益，也是为了为其他领域的研究人员提供开箱即用的 SLAM 解决方案。

一、简介

同步定位与建图 (SLAM) 是过去二十年计算机视觉和机器人领域的热门研究课题，最近也引起了高科技公司的关注。SLAM 技术构建未知环境的地图，并在地图中定位传感器，重点关注实时操作。在不同的传感器模式中，相机价格便宜，并提供丰富的环境信息，可以实现稳健而准确的地点识别。因此，以摄像头为主要传感器的视觉 SLAM 解决方案如今备受关注。位置识别是 SLAM 系统的关键模块，用于闭环（即检测传感器何时返回映射区域并纠正探索中累积的误差），并在由于遮挡或攻击性运动而导致跟踪失败后或在系统重新初始化时重新定位相机。

视觉 SLAM 可以仅使用单目相机来执行，这是最便宜和最小的传感器设置。然而，由于仅通过一台摄像机无法观察到深度，因此地图的比例和估计轨迹是未知的。此外，系统引导需要多视图或过滤技术来生成初始地图，因为它无法从第一帧开始进行三角测量。最后但并非最不重要的

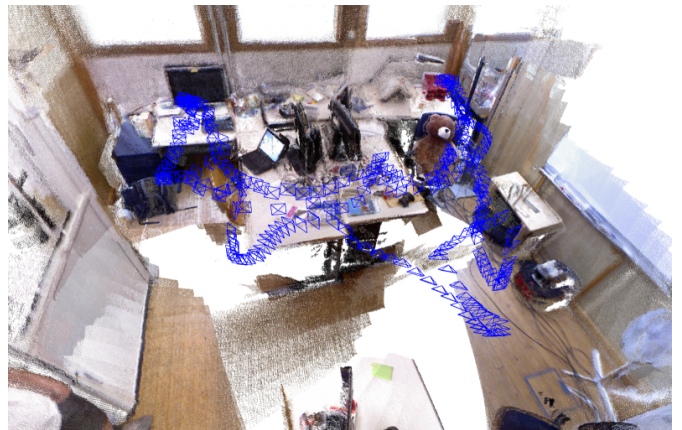
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(a) 立体输入：具有多个闭环的城市环境的轨迹和稀疏重建。



(b) RGB-D 输入：具有一个循环闭合的房间场景的关键帧和密集点云。点云是通过从估计的关键帧姿势反投影传感器深度图来渲染的。不进行融合。

图 1. ORB-SLAM2 处理立体和 RGB-D 输入以估计相机轨迹并构建环境地图。该系统能够在标准 CPU 上实时闭环、重新定位和重用其地图，具有高精度和鲁棒性。

单目 SLAM 会受到尺度漂移的影响，如果在探索中执行纯旋转，则可能会失败。通过使用立体或 RGB-D 相机，所有这些问题都得到解决，并允许最可靠的视觉 SLAM 解决方案。

在本文中，我们在单目 ORB-SLAM [1] 的基础上提出了 ORB-SLAM2，其贡献如下：

- The first open-source¹ SLAM system for monocular, stereo and RGB-D cameras, including loop closing, relocalization and map reuse.
- Our RGB-D results show that by using Bundle Adjustment (BA) we achieve more accuracy than state-of-the-art methods based on ICP or photometric and depth error minimization.
- By using close and far stereo points and monocular observations our stereo results are more accurate than the state-of-the-art direct stereo SLAM.
- A lightweight localization mode that can effectively reuse the map with mapping disabled.

Fig. 1 shows examples of ORB-SLAM2 output from stereo and RGB-D inputs. The stereo case shows the final trajectory and sparse reconstruction of the sequence 00 from the KITTI dataset [2]. This is an urban sequence with multiple loop closures that ORB-SLAM2 was able to successfully detect. The RGB-D case shows the keyframe poses estimated in sequence fr1_room from the TUM RGB-D Dataset [3], and a dense pointcloud, rendered by backprojecting sensor depth maps from the estimated keyframe poses. Note that our SLAM does not perform any fusion like KinectFusion [4] or similar, but the good definition indicates the accuracy of the keyframe poses. More examples are shown on the attached video.

In the rest of the paper, we discuss related work in Section II, we describe our system in Section III, then present the evaluation results in Section IV and end with conclusions in Section V.

II. RELATED WORK

In this section we discuss related work on stereo and RGB-D SLAM. Our discussion, as well as the evaluation in Section IV is focused only on SLAM approaches.

A. Stereo SLAM

A remarkable early stereo SLAM system was the work of Paz et al. [5]. Based on Conditionally Independent Divide and Conquer EKF-SLAM it was able to operate in larger environments than other approaches at that time. Most importantly, it was the first stereo SLAM exploiting both close and far points (i.e. points whose depth cannot be reliably estimated due to little disparity in the stereo camera), using an inverse depth parametrization [6] for the latter. They empirically showed that points can be reliably triangulated if their depth is less than ~ 40 times the stereo baseline. In this work we follow this strategy of treating in a different way *close* and *far* points, as explained in Section III-A.

Most modern stereo SLAM systems are keyframe-based [7] and perform BA optimization in a local area to achieve scalability. The work of Strasdat et al. [8] performs a joint optimization of BA (point-pose constraints) in an inner window of keyframes and pose-graph (pose-pose constraints) in an outer window. By limiting the size of these windows the method achieves constant time complexity, at the expense of not guaranteeing global consistency. The RSLAM of Mei et

al. [9] uses a relative representation of landmarks and poses and performs relative BA in an active area which can be constrained for constant-time. RSLAM is able to close loops which allow to expand active areas at both sides of a loop, but global consistency is not enforced. The recent S-PTAM by Pire et al. [10] performs local BA, however it lacks large loop closing. Similar to these approaches we perform BA in a local set of keyframes so that the complexity is independent of the map size and we can operate in large environments. However our goal is to build a globally consistent map. When closing a loop, our system aligns first both sides, similar to RSLAM, so that the tracking is able to continue localizing using the old map and then performs a pose-graph optimization that minimizes the drift accumulated in the loop, followed by full BA.

The recent Stereo LSD-SLAM of Engel et al. [11] is a semi-dense direct approach that minimizes photometric error in image regions with high gradient. Not relying on features, the method is expected to be more robust to motion blur or poorly-textured environments. However as a direct method its performance can be severely degraded by unmodeled effects like rolling shutter or non-lambertian reflectance.

B. RGB-D SLAM

One of the earliest and most famed RGB-D SLAM systems was the KinectFusion of Newcombe et al. [4]. This method fused all depth data from the sensor into a volumetric dense model that is used to track the camera pose using ICP. This system was limited to small workspaces due to its volumetric representation and the lack of loop closing. Kintinuous by Whelan et al. [12] was able to operate in large environments by using a rolling cyclical buffer and included loop closing using place recognition and pose graph optimization.

Probably the first popular open-source system was the RGB-D SLAM of Endres et al. [13]. This is a feature-based system, whose front-end computes frame-to-frame motion by feature matching and ICP. The back-end performs pose-graph optimization with loop closure constraints from a heuristic search. Similarly the back-end of DVO-SLAM by Kerl et al. [14] optimizes a pose-graph where keyframe-to-keyframe constraints are computed from a visual odometry that minimizes both photometric and depth error. DVO-SLAM also searches for loop candidates in a heuristic fashion over all previous frames, instead of relying on place recognition.

The recent ElasticFusion of Whelan et al. [15] builds a surfel-based map of the environment. This is a map-centric approach that forget poses and performs loop closing applying a non-rigid deformation to the map, instead of a standard pose-graph optimization. The detailed reconstruction and localization accuracy of this system is impressive, but the current implementation is limited to room-size maps as the complexity scales with the number of surfels in the map.

As proposed by Strasdat et al. [8] our ORB-SLAM2 uses depth information to synthesize a stereo coordinate for extracted features on the image. This way our system is agnostic of the input being stereo or RGB-D. Differently to all above methods our back-end is based on bundle adjustment and

¹https://github.com/raulmur/ORB_SLAM2

- 第一个用于单目、立体和 RGB-D 相机的开源¹ SLAM 系统，包括闭环、重新定位和地图重用。
- 我们的 RGB-D 结果表明，通过使用束调整 (BA)，我们比基于 ICP 或光度和深度误差最小化的最先进方法获得了更高的精度。
- 通过使用近距离和远距离立体点以及单目观察，我们的立体结果比最先进的直接立体 SLAM 更准确。
- 一种轻量级的定位模式，可以在禁用映射的情况下有效地重用地图。

图 1 显示了立体和 RGB-D 输入的 ORB-SLAM2 输出示例。立体案例显示了 KITTI 数据集 [2] 中序列 00 的最终轨迹和稀疏重建。这是 ORB-SLAM2 能够成功检测到的具有多个闭环的城市序列。RGB-D 案例显示了根据 TUM RGB-D 数据集 [3] 在序列 fr1 room 中估计的关键帧姿势，以及通过从估计的关键帧姿势反投影传感器深度图来渲染的密集点云。请注意，我们的 SLAM 不会像 KinectFusion [4] 或类似的那样执行任何融合，但良好的定义表明了关键帧姿势的准确性。附加视频中显示了更多示例。

在本文的其余部分，我们在第二节中讨论相关工作，在第三节中描述我们的系统，然后在第四节中呈现评估结果，并在第五节中得出结论。

二.相关工作

在本节中，我们将讨论立体和 RGB-D SLAM 的相关工作。我们的讨论以及第四节中的评估仅集中于 SLAM 方法。

A. Stereo SLAM

Paz 等人的工作是一个卓越的早期立体 SLAM 系统。[5]。基于条件独立分而治之的 EKF-SLAM，它能够在比当时其他方法更大的环境中运行。最重要的是，这是第一个同时利用近点和远点（即由于立体相机差异很小而无法可靠估计深度的点）的立体 SLAM，对后者使用逆深度参数化 [6]。他们凭经验表明，如果点的深度小于立体基线的 ~ 40 倍，则可以可靠地对点进行三角测量。在这项工作中，我们遵循以不同方式处理 *close* 和 *far* 点的策略，如第 III-A 节中所述。

大多数现代立体 SLAM 系统都是基于关键帧的 [7]，并在局部区域执行 BA 优化以实现可扩展性。Strasdat 等人的工作。[8]在关键帧的内部窗口中执行 BA（点位姿约束）和外部窗口中的位姿图（位姿约束）的联合优化。通过限制这些窗口的大小，该方法实现了恒定的时间复杂度，但代价是不保证全局一致性。Mei 等人的 RSLAM

等人。[9]使用地标和姿势的相对表示，并在活动区域中执行相对 BA，该活动区域可以被限制为恒定时间。RSLAM 能够闭合循环，从而允许扩展循环两侧的活动区域，但不强制执行全局一致性。Pire 等人最近提出的 S-PTAM。[10]执行局部 BA，但是它缺乏大的循环闭合。与这些方法类似，我们在一组本地关键帧中执行 BA，以便复杂性与地图大小无关，并且我们可以在大型环境中操作。然而我们的目标是建立一个全球一致的地图。当闭合循环时，我们的系统首先对齐两侧，类似于 RSLAM，以便跟踪能够使用旧地图继续定位，然后执行位姿图优化，最大限度地减少循环中累积的漂移，然后进行完整的 BA。

Engel 等人最近提出的 Stereo LSD-SLAM。[11]是一种半密集直接方法，可以最大限度地减少高梯度图像区域的光度误差。该方法不依赖于特征，预计对于运动模糊或纹理不良的环境具有更强的鲁棒性。然而，作为一种直接方法，其性能可能会因卷帘快门或非朗伯反射等未建模效应而严重降低。

B. RGB-D SLAM

Newcombe 等人的 KinectFusion 是最早、最著名的 RGB-D SLAM 系统之一。[4]。该方法将来自传感器的所有深度数据融合到体积密集模型中，该模型用于使用 ICP 跟踪相机姿态。由于其体积表示和缺乏循环闭合，该系统仅限于小型工作空间。惠兰 (Whelan) 等人的《Kintinuous》。[12]能够通过使用滚动循环缓冲区在大型环境中运行，并包括使用位置识别和位姿图优化的循环闭合。

第一个流行的开源系统可能是 Endres 等人的 RGB-D SLAM。[13]。这是一个基于特征的系统，其前端通过特征匹配和 ICP 来计算帧到帧的运动。后端通过启发式搜索的闭环约束执行位姿图优化。Kerl 等人的 DVO-SLAM 后端也类似。[14]优化了姿态图，其中关键帧到关键帧的约束是根据视觉里程计计算的，从而最大限度地减少了光度和深度误差。DVO-SLAM 还以启发式方式在所有先前帧中搜索循环候选，而不是依赖位置识别。

Whelan 等人最近的 ElasticFusion。[15]构建了一个基于面元的环境地图。这是一种以地图为中心的方法，它会忘记姿势并执行循环闭合，对地图应用非刚性变形，而不是标准的姿势图优化。该系统的详细重建和定位精度令人印象深刻，但当前的实现仅限于房间大小的地图，因为复杂性随着地图中面元的数量而变化。

正如 Strasdat 等人提出的。[8]我们的 ORB-SLAM2 使用深度信息来合成图像上提取的特征的立体坐标。这样我们的系统就不会知道输入是立体声还是 RGB-D。与上述所有方法不同，我们的后端基于捆绑调整和

¹https://github.com/raulmur/ORB_SLAM2

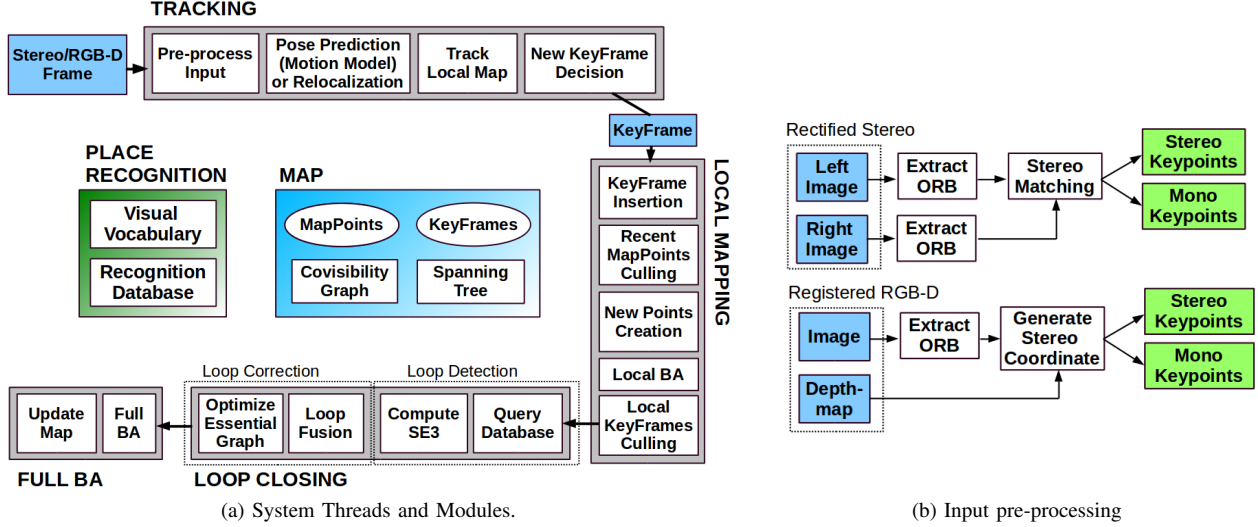


Fig. 2. ORB-SLAM2 is composed of three main parallel threads: tracking, local mapping and loop closing, which can create a fourth thread to perform full BA after a loop closure. The tracking thread pre-processes the stereo or RGB-D input so that the rest of the system operates independently of the input sensor. Although it is not shown in this figure, ORB-SLAM2 also works with a monocular input as in [1].

builds a globally consistent sparse reconstruction. Therefore our method is lightweight and works with standard CPUs. Our goal is long-term and globally consistent localization instead of building the most detailed dense reconstruction. However from the highly accurate keyframe poses one could fuse depth maps and get accurate reconstruction on-the-fly in a local area or post-process the depth maps from all keyframes after a full BA and get an accurate 3D model of the whole scene.

III. ORB-SLAM2

ORB-SLAM2 for stereo and RGB-D cameras is built on our monocular feature-based ORB-SLAM [1], whose main components are summarized here for reader convenience. A general overview of the system is shown in Fig. 2. The system has three main parallel threads: 1) the tracking to localize the camera with every frame by finding feature matches to the local map and minimizing the reprojection error applying motion-only BA, 2) the local mapping to manage the local map and optimize it, performing local BA, 3) the loop closing to detect large loops and correct the accumulated drift by performing a pose-graph optimization. This thread launches a fourth thread to perform full BA after the pose-graph optimization, to compute the optimal structure and motion solution.

The system has embedded a Place Recognition module based on DBoW2 [16] for relocalization, in case of tracking failure (e.g. an occlusion) or for reinitialization in an already mapped scene, and for loop detection. The system maintains a covisibility graph [8] that links any two keyframes observing common points and a minimum spanning tree connecting all keyframes. These graph structures allow to retrieve local windows of keyframes, so that tracking and local mapping operate locally, allowing to work on large environments, and serve as structure for the pose-graph optimization performed when closing a loop.

The system uses the same ORB features [17] for tracking, mapping and place recognition tasks. These features are robust

to rotation and scale and present a good invariance to camera auto-gain and auto-exposure, and illumination changes. Moreover they are fast to extract and match allowing for real-time operation and show good precision/recall performance in bag-of-words place recognition [18].

In the rest of this section we present how stereo/depth information is exploited and which elements of the system are affected. For a detailed description of each system block, we refer the reader to our monocular publication [1].

A. Monocular, Close Stereo and Far Stereo Keypoints

ORB-SLAM2 as a feature-based method pre-processes the input to extract features at salient keypoint locations, as shown in Fig. 2b. The input images are then discarded and all system operations are based on these features, so that the system is independent of the sensor being stereo or RGB-D. Our system handles monocular and stereo keypoints, which are further classified as close or far.

Stereo keypoints are defined by three coordinates $\mathbf{x}_s = (u_L, v_L, u_R)$, being (u_L, v_L) the coordinates on the left image and u_R the horizontal coordinate in the right image. For stereo cameras, we extract ORB in both images and for every left ORB we search for a match in the right image. This can be done very efficiently assuming stereo rectified images, so that epipolar lines are horizontal. We then generate the stereo keypoint with the coordinates of the left ORB and the horizontal coordinate of the right match, which is subpixel refined by patch correlation. For RGB-D cameras, we extract ORB features on the RGB image and, as proposed by Strasdat et al. [8], for each feature with coordinates (u_L, v_L) we transform its depth value d into a virtual right coordinate:

$$u_R = u_L - \frac{f_x b}{d} \quad (1)$$

where f_x is the horizontal focal length and b is the baseline between the structured light projector and the infrared camera, which we approximate to 8cm for Kinect and Asus Xtion.

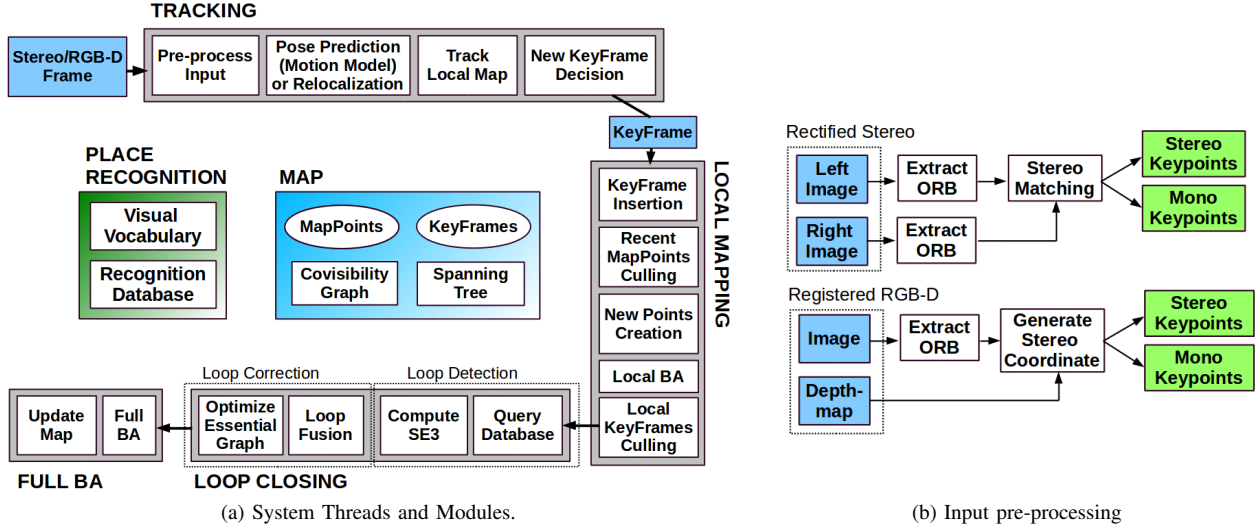


图2. ORB-SLAM2由三个主要并行线程组成：跟踪、局部建图和闭环，在闭环后可以创建第四个线程来执行完整的BA。跟踪线程对立体声或 RGB-D 输入进行预处理，以便系统的其余部分独立于输入传感器运行。尽管图中未显示，ORB-SLAM2 也可以使用单目输入，如 [1] 中所示。

构建全局一致的稀疏重建。因此，我们的方法是轻量级的，并且适用于标准 CPU。我们的目标是长期且全球一致的定位，而不是构建最详细的密集重建。然而，从高度准确的关键帧姿势中，我们可以融合深度图并在局部区域中即时获得准确的重建，或者在完整的 BA 后对所有关键帧的深度图进行后处理，并获得整个场景的准确 3D 模型。

三. ORB-SLAM2

用于立体和 RGB-D 相机的 ORB-SLAM2 建立在我们基于单目特征的 ORB-SLAM [1] 的基础上，为了方便读者，这里总结了其主要组件。该系统的总体概述如图 2 所示。该系统具有三个主要并行线程：1) 通过查找与局部图匹配的特征并最小化应用仅运动 BA 的重投影误差来对每一帧进行跟踪以定位相机，2) 局部映射以管理局部图并对其进行优化，执行局部 BA，3) 循环闭合以检测大循环并通过执行位姿图优化来校正累积漂移。该线程在位姿图优化之后启动第四个线程来执行完整的 BA，以计算最佳结构和运动解决方案。

该系统嵌入了基于 DBoW2 [16] 的位置识别模块，用于在跟踪失败（例如遮挡）的情况下重新定位或在已映射的场景中重新初始化，以及用于循环检测。该系统维护一个可观察图[8]，它链接观察公共点的任意两个关键帧和连接所有关键帧的最小生成树。这些图结构允许检索关键帧的本地窗口，以便跟踪和本地映射在本地运行，允许在大型环境中工作，并用作闭合循环时执行的位姿图优化的结构。

该系统使用相同的 ORB 功能 [17] 来执行跟踪、绘图和地点识别任务。这些功能很强大

旋转和缩放，并对相机自动增益和自动曝光以及照明变化呈现良好的不变性。此外，它们可以快速提取和匹配，允许实时操作，并在词袋位置识别中表现出良好的精度/召回性能[18]。

在本节的其余部分中，我们将介绍如何利用立体/深度信息以及系统的哪些元素受到影响。有关每个系统块的详细描述，我们建议读者参阅我们的单目出版物 [1]。

A. Monocular, Close Stereo and Far Stereo Keypoints

ORB-SLAM2作为一种基于特征的方法对输入进行预处理以提取显著关键点位置的特征，如图2b所示。然后输入图像被丢弃，所有系统操作都基于这些特征，因此系统独立于传感器是立体还是RGB-D。我们的系统处理单眼和立体关键点，这些关键点进一步分为近点和远点。

立体关键点由三个坐标 $\mathbf{x}_s = (u_L, v_L, u_R)$ 定义，即 (u_L, v_L) 是左图像上的坐标， u_R 是右图像中的水平坐标。对于立体相机，我们在两个图像中提取 ORB，并且对于每个左侧 ORB，我们在右侧图像中搜索匹配项。假设立体校正图像可以非常有效地完成此操作，因此极线是水平的。然后，我们使用左侧 ORB 的坐标和右侧匹配的水平坐标生成立体关键点，该关键点是通过补丁相关性细化的子像素。对于 RGB-D 相机，我们在 RGB 图像上提取 ORB 特征，并且如 Strasdat 等人提出的那样。[8]，对于每个具有坐标 (u_L, v_L) 的特征，我们将其深度值 d 转换为虚拟右坐标：

$$u_R = u_L - \frac{f_x b}{d} \quad (1)$$

其中 f_x 是水平焦距， b 是结构光投影仪和红外摄像头之间的基线，对于 Kinect 和 Asus Xtion，我们将其近似为 8 厘米。

The uncertainty of the depth sensor is represented by the uncertainty of the virtual right coordinate. In this way, features from stereo and RGB-D input are handled equally by the rest of the system.

A stereo keypoint is classified as **close** if its associated depth is less than 40 times the stereo/RGB-D baseline, as suggested in [5], otherwise it is classified as **far**. Close keypoints can be safely triangulated from one frame as depth is accurately estimated and provide scale, translation and rotation information. On the other hand far points provide accurate rotation information but weaker scale and translation information. We triangulate far points when they are supported by multiple views.

Monocular keypoints are defined by two coordinates $\mathbf{x}_m = (u_L, v_L)$ on the left image and correspond to all those ORB for which a stereo match could not be found or that have an invalid depth value in the RGB-D case. These points are only triangulated from multiple views and do not provide scale information, but contribute to the rotation and translation estimation.

B. System Bootstrapping

One of the main benefits of using stereo or RGB-D cameras is that, by having depth information from just one frame, we do not need a specific structure from motion initialization as in the monocular case. At system startup we create a keyframe with the first frame, set its pose to the origin, and create an initial map from all stereo keypoints.

C. Bundle Adjustment with Monocular and Stereo Constraints

Our system performs BA to optimize the camera pose in the tracking thread (motion-only BA), to optimize a local window of keyframes and points in the local mapping thread (local BA), and after a loop closure to optimize all keyframes and points (full BA). We use the Levenberg–Marquardt method implemented in g2o [19].

Motion-only BA optimizes the camera orientation $\mathbf{R} \in SO(3)$ and position $\mathbf{t} \in \mathbb{R}^3$, minimizing the reprojection error between matched 3D points $\mathbf{X}^i \in \mathbb{R}^3$ in world coordinates and keypoints $\mathbf{x}_{(\cdot)}^i$, either monocular $\mathbf{x}_m^i \in \mathbb{R}^2$ or stereo $\mathbf{x}_s^i \in \mathbb{R}^3$, with $i \in \mathcal{X}$ the set of all matches:

$$\{\mathbf{R}, \mathbf{t}\} = \underset{\mathbf{R}, \mathbf{t}}{\operatorname{argmin}} \sum_{i \in \mathcal{X}} \rho \left(\left\| \mathbf{x}_{(\cdot)}^i - \pi_{(\cdot)}(\mathbf{R}\mathbf{X}^i + \mathbf{t}) \right\|_{\Sigma}^2 \right) \quad (2)$$

where ρ is the robust Huber cost function and Σ the covariance matrix associated to the scale of the keypoint. The projection functions $\pi_{(\cdot)}$, monocular π_m and rectified stereo π_s , are defined as follows:

$$\pi_m \left(\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \right) = \begin{bmatrix} f_x \frac{X}{Z} + c_x \\ f_y \frac{Y}{Z} + c_y \end{bmatrix}, \pi_s \left(\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \right) = \begin{bmatrix} f_x \frac{X}{Z} + c_x \\ f_y \frac{Y}{Z} + c_y \\ f_x \frac{X-b}{Z} + c_x \end{bmatrix} \quad (3)$$

where (f_x, f_y) is the focal length, (c_x, c_y) is the principal point and b the baseline, all known from calibration.

Local BA optimizes a set of covisible keyframes $\mathcal{K}_L$ and all points seen in those keyframes $\mathcal{P}_L$. All other keyframes $\mathcal{K}_F$, not in $\mathcal{K}_L$, observing points in $\mathcal{P}_L$ contribute to the cost function but remain fixed in the optimization. Defining $\mathcal{X}_k$ as the set of matches between points in $\mathcal{P}_L$ and keypoints in a keyframe k , the optimization problem is the following:

$$\{\mathbf{X}^i, \mathbf{R}_l, \mathbf{t}_l | i \in \mathcal{P}_L, l \in \mathcal{K}_L\} = \underset{\mathbf{X}^i, \mathbf{R}_l, \mathbf{t}_l}{\operatorname{argmin}} \sum_{k \in \mathcal{K}_L \cup \mathcal{K}_F} \sum_{j \in \mathcal{X}_k} \rho(E_{kj})$$

$$E_{kj} = \left\| \mathbf{x}_{(\cdot)}^j - \pi_{(\cdot)}(\mathbf{R}_k \mathbf{X}^j + \mathbf{t}_k) \right\|_{\Sigma}^2 \quad (4)$$

Full BA is the specific case of local BA, where all keyframes and points in the map are optimized, except the origin keyframe that is fixed to eliminate the gauge freedom.

D. Loop Closing and Full BA

Loop closing is performed in two steps, firstly a loop has to be detected and validated, and secondly the loop is corrected optimizing a pose-graph. In contrast to monocular ORB-SLAM, where scale drift may occur [20], the stereo/depth information makes scale observable and the geometric validation and pose-graph optimization no longer require dealing with scale drift and are based on rigid body transformations instead of similarities.

In ORB-SLAM2 we have incorporated a full BA optimization after the pose-graph to achieve the optimal solution. This optimization might be very costly and therefore we perform it in a separate thread, allowing the system to continue creating map and detecting loops. However this brings the challenge of merging the bundle adjustment output with the current state of the map. If a new loop is detected while the optimization is running, we abort the optimization and proceed to close the loop, which will launch the full BA optimization again. When the full BA finishes, we need to merge the updated subset of keyframes and points optimized by the full BA, with the non-updated keyframes and points that were inserted while the optimization was running. This is done by propagating the correction of updated keyframes (i.e. the transformation from the non-optimized to the optimized pose) to non-updated keyframes through the spanning tree. Non-updated points are transformed according to the correction applied to their reference keyframe.

E. Keyframe Insertion

ORB-SLAM2 follows the policy introduced in monocular ORB-SLAM of inserting keyframes very often and culling redundant ones afterwards. The distinction between close and far stereo points allows us to introduce a new condition for keyframe insertion, which can be critical in challenging environments where a big part of the scene is far from the stereo sensor, as shown in Fig. 3. In such environment we need to have a sufficient amount of close points to accurately estimate translation, therefore if the number of tracked close points drops below τ_t and the frame could create at least τ_c new close stereo points, the system will insert a new keyframe. We empirically found that $\tau_t = 100$ and $\tau_c = 70$ works well in all our experiments.

深度传感器的不确定度用虚拟右坐标的不确定度来表示。通过这种方式，系统的其余部分可以同等地处理来自立体声和 RGB-D 输入的功能。

如果立体关键点的关联深度小于立体/RGB-D 基线的 40 倍（如 [5] 中所建议的），则该立体关键点被分类为近，否则它被分类为远。由于深度被准确估计并提供缩放、平移和旋转信息，因此可以从一帧安全地对闭合关键点进行三角测量。另一方面，远点提供准确的旋转信息，但缩放和平移信息较弱。当远点受到多个视图支持时，我们会对远点进行三角测量。

单目关键点由左侧图像上的两个坐标 $\mathbf{x}_m = (u_L, v_L)$ 定义，并对应于无法找到立体匹配或在 RGB-D 情况下具有无效深度值的所有 ORB。这些点仅从多个视图进行三角测量，不提供比例信息，但有助于旋转和平移估计。

B. System Bootstrapping

使用立体或 RGB-D 相机的主要好处之一是，通过仅从一帧获得深度信息，我们不需要像单目情况那样从运动初始化中获得特定的结构。在系统启动时，我们使用第一帧创建一个关键帧，将其姿势设置为原点，并从所有立体关键点创建一个初始贴图。

C. Bundle Adjustment with Monocular and Stereo Constraints

我们的系统执行 BA 来优化跟踪线程中的相机姿态（仅运动 BA），优化本地映射线程中的关键帧和点的本地窗口（本地 BA），并在循环闭合后优化所有关键帧和点（完整 BA）。我们使用 g2o [19] 中实现的 Levenberg-Marquardt 方法。

仅运动 BA 优化相机方向 $\mathbf{R} \in SO(3)$ 和位置 $\mathbf{t} \in \mathbb{R}^3$ ，最小化世界坐标中匹配的 3D 点 $\mathbf{X}^i \in \mathbb{R}^3$ 和关键点 $\mathbf{x}_{(\cdot)}^i$ （单目 $\mathbf{x}_m^i \in \mathbb{R}^2$ 或立体 $\mathbf{x}_s^i \in \mathbb{R}^3$ ）之间的重投影误差，其中 $i \in \mathcal{X}$ 是所有匹配的集合：

$$\{\mathbf{R}, \mathbf{t}\} = \underset{\mathbf{R}, \mathbf{t}}{\operatorname{argmin}} \sum_{i \in \mathcal{X}} \rho \left(\left\| \mathbf{x}_{(\cdot)}^i - \pi_{(\cdot)}(\mathbf{R}\mathbf{X}^i + \mathbf{t}) \right\|_{\Sigma}^2 \right) \quad (2)$$

其中 ρ 是稳健的 Huber 成本函数， Σ 是与关键点尺度相关的协方差矩阵。投影函数 $\pi_{(\cdot)}$ 、单目投影函数 π_m 和矫正立体投影函数 π_s 定义如下：

$$\pi_m \left(\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \right) = \begin{bmatrix} f_x \frac{X}{Z} + c_x \\ f_y \frac{Y}{Z} + c_y \end{bmatrix}, \pi_s \left(\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \right) = \begin{bmatrix} f_x \frac{X}{Z} + c_x \\ f_y \frac{Y}{Z} + c_y \\ f_x \frac{X-b}{Z} + c_x \end{bmatrix} \quad (3)$$

其中 (f_x, f_y) 是焦距， (c_x, c_y) 是主点， b 是基线，所有这些都可以通过校准得知。

本地 BA 优化一组可见关键帧 $\mathcal{K}_L$ 以及这些关键帧 $\mathcal{P}_L$ 中看到的所有点。所有其他关键帧 $\mathcal{K}_F$ （不在 $\mathcal{K}_L$ 中）、 $\mathcal{P}_L$ 中的观察点都对成本函数有贡献，但在优化中保持固定。将 $\mathcal{X}_k$ 定义为 $\mathcal{P}_L$ 中的点与关键帧 k 中的关键点之间的匹配集，优化问题如下：

$$\{\mathbf{X}^i, \mathbf{R}_l, \mathbf{t}_l | i \in \mathcal{P}_L, l \in \mathcal{K}_L\} = \underset{\mathbf{X}^i, \mathbf{R}_l, \mathbf{t}_l}{\operatorname{argmin}} \sum_{k \in \mathcal{K}_L \cup \mathcal{K}_F} \sum_{j \in \mathcal{X}_k} \rho(E_{kj})$$

$$E_{kj} = \left\| \mathbf{x}_{(\cdot)}^j - \pi_{(\cdot)}(\mathbf{R}_k \mathbf{X}^j + \mathbf{t}_k) \right\|_{\Sigma}^2 \quad (4)$$

完整 BA 是局部 BA 的特例，其中地图中的所有关键帧和点都经过优化，但原始关键帧除外，该关键帧被固定以消除仪表自由度。

D. Loop Closing and Full BA

循环闭合分两个步骤执行，首先必须检测和验证循环，其次通过优化姿态图来校正循环。与可能发生尺度漂移的单目 ORB-SLAM 相比 [20]，立体/深度信息使得尺度可观察，并且几何验证和位姿图优化不再需要处理尺度漂移，并且基于刚体变换而不是相似性。

在 ORB-SLAM2 中，我们在位姿图之后结合了完整的 BA 优化，以实现最佳解决方案。这种优化的成本可能非常高，因此我们在单独的线程中执行它，允许系统继续创建地图和检测循环。然而，这带来了将捆绑调整输出与地图的当前状态合并的挑战。如果在优化运行时检测到新循环，我们将中止优化并继续关闭循环，这将再次启动完整的 BA 优化。当完整的 BA 完成时，我们需要将完整 BA 优化的关键帧和点的更新子集与优化运行时插入的未更新的关键帧和点合并。这是通过通过生成树将更新关键帧的校正（即从非优化姿势到优化姿势的变换）传播到非更新关键帧来完成的。未更新的点根据应用于其参考关键帧的校正进行变换。

E. Keyframe Insertion

ORB-SLAM2 遵循单目 ORB-SLAM 中引入的频繁插入关键帧并随后剔除冗余帧的策略。近点和远点立体点之间的区别使我们能够引入关键帧插入的新条件，这在场景的大部分远离立体传感器的具有挑战性的环境中至关重要，如图 3 所示。在这种环境中，我们需要有足够数量的近点来准确估计平移，因此，如果跟踪的近点数量下降低于 τ_t ，并且该帧可以创建至少 τ_c 个新的近点立体点，系统将插入新的关键帧。我们凭经验发现 $\tau_t = 100$ 和 $\tau_c = 70$ 在我们所有的实验中效果良好。

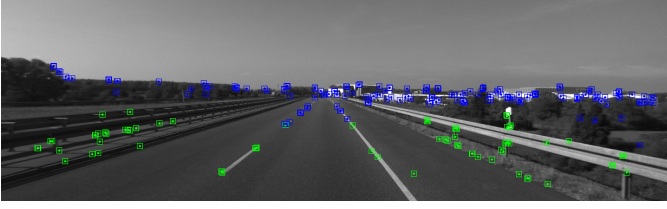


Fig. 3. Tracked points in KITTI 01 [2]. Green points have a depth less than 40 times the stereo baseline, while blue points are further away. In this kind of sequences it is important to insert keyframes often enough so that the amount of close points allows for accurate translation estimation. Far points contribute to estimate orientation but provide weak information for translation and scale.

F. Localization Mode

We incorporate a Localization Mode which can be useful for lightweight long-term localization in well mapped areas, as long as there are not significant changes in the environment. In this mode the local mapping and loop closing threads are deactivated and the camera is continuously localized by the tracking using relocalization if needed. In this mode the tracking leverages visual odometry matches and matches to map points. Visual odometry matches are matches between ORB in the current frame and 3D points created in the previous frame from the stereo/depth information. These matches make the localization robust to unmapped regions, but drift can be accumulated. Map point matches ensure drift-free localization to the existing map. This mode is demonstrated in the accompanying video.

IV. EVALUATION

We have evaluated ORB-SLAM2 in three popular datasets and compared to other state-of-the-art SLAM systems, using always the results published by the original authors and standard evaluation metrics in the literature. We have run ORB-SLAM2 in an Intel Core i7-4790 desktop computer with 16Gb RAM. In order to account for the non-deterministic nature of the multi-threading system, we run each sequence 5 times and show median results for the accuracy of the estimated trajectory. Our open-source implementation includes the calibration and instructions to run the system in all these datasets.

A. KITTI Dataset

The KITTI dataset [2] contains stereo sequences recorded from a car in urban and highway environments. The stereo sensor has a $\sim 54\text{cm}$ baseline and works at 10Hz with a resolution after rectification of 1240×376 pixels. Sequences 00, 02, 05, 06, 07 and 09 contain loops. Our ORB-SLAM2 detects all loops and is able to reuse its map afterwards, except for sequence 09 where the loop happens in very few frames at the end of the sequence. Table I shows results in the 11 training sequences, which have public ground-truth, compared to the state-of-the-art Stereo LSD-SLAM [11], to our knowledge the only stereo SLAM showing detailed results for all sequences. We use two different metrics, the absolute translation RMSE t_{abs} proposed in [3], and the average relative translation t_{rel} and rotation r_{rel} errors proposed in [2]. Our

TABLE I
COMPARISON OF ACCURACY IN THE KITTI DATASET.

Sequence	ORB-SLAM2 (stereo)			Stereo LSD-SLAM		
	t_{rel} (%)	r_{rel} (deg/100m)	t_{abs} (m)	t_{rel} (%)	r_{rel} (deg/100m)	t_{abs} (m)
00	0.70	0.25	1.3	0.63	0.26	1.0
01	1.39	0.21	10.4	2.36	0.36	9.0
02	0.76	0.23	5.7	0.79	0.23	2.6
03	0.71	0.18	0.6	1.01	0.28	1.2
04	0.48	0.13	0.2	0.38	0.31	0.2
05	0.40	0.16	0.8	0.64	0.18	1.5
06	0.51	0.15	0.8	0.71	0.18	1.3
07	0.50	0.28	0.5	0.56	0.29	0.5
08	1.05	0.32	3.6	1.11	0.31	3.9
09	0.87	0.27	3.2	1.14	0.25	5.6
10	0.60	0.27	1.0	0.72	0.33	1.5

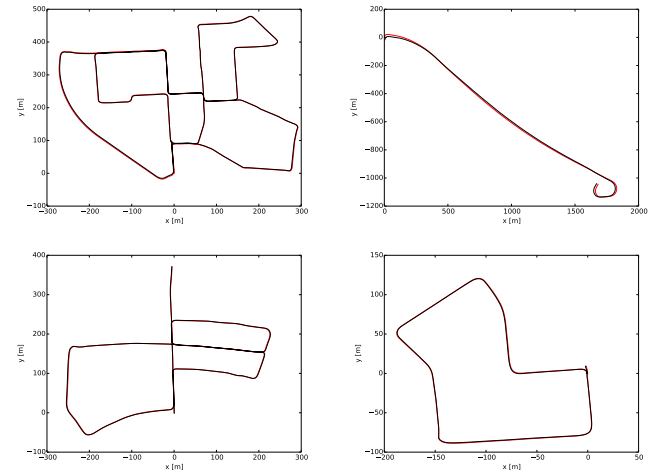


Fig. 4. Estimated trajectory (black) and ground-truth (red) in KITTI 00, 01, 05 and 07.

system outperforms Stereo LSD-SLAM in most sequences, and achieves in general a relative error lower than 1%. The sequence 01, see Fig. 3, is the only highway sequence in the training set and the translation error is slightly worse. Translation is harder to estimate in this sequence because very few close points can be tracked, due to high speed and low frame-rate. However orientation can be accurately estimated, achieving an error of 0.21 degrees per 100 meters, as there are many far point that can be long tracked. Fig. 4 shows some examples of estimated trajectories.

Compared to the monocular results presented in [1], the proposed stereo version is able to process the sequence 01 where the monocular system failed. In this highway sequence, see Fig. 3, close points are in view only for a few frames. The ability of the stereo version to create points from just one stereo keyframe instead of the delayed initialization of the monocular, consisting on finding matches between two keyframes, is critical in this sequence not to lose tracking. Moreover the stereo system estimates the map and trajectory with metric scale and does not suffer from scale drift, as seen in Fig. 5.

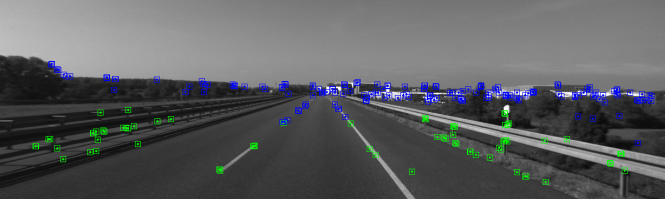


图 3. KITTI 01 [2] 中的跟踪点。绿点的深度小于立体基线的 40 倍，而蓝点距离更远。在这种序列中，重要的是经常插入关键帧，以便接近点的数量可以进行准确的平移估计。远点有助于估计方向，但为平移和缩放提供的信息较弱。

F. Localization Mode

我们采用了一种定位模式，只要环境没有重大变化，该模式对于地图绘制良好的区域中的轻量级长期定位很有用。在此模式下，局部映射和循环闭合线程将被停用，并且如果需要，可以通过使用重新定位的跟踪来连续定位相机。在此模式下，跟踪利用视觉里程计匹配和地图点匹配。视觉里程计匹配是当前帧中的 ORB 与根据立体/深度信息在前一帧中创建的 3D 点之间的匹配。这些匹配使得定位对未映射的区域具有鲁棒性，但漂移可能会累积。地图点匹配可确保对现有地图的无漂移定位。随附的视频演示了此模式。

四. 评估

我们在三个流行的数据集中评估了 ORB-SLAM2，并与其他最先进的 SLAM 系统进行了比较，始终使用原作者发布的结果和文献中的标准评估指标。我们在具有 16Gb RAM 的 Intel Core i7-4790 台式计算机上运行 ORB-SLAM2。为了考虑多线程系统的不确定性，我们运行每个序列 5 次，并显示估计轨迹准确性的中值结果。我们的开源实现包括在所有这些数据集中运行系统的校准和说明。

A. KITTI Dataset

KITTI 数据集 [2] 包含从城市和高速公路环境中的汽车记录的立体声序列。立体传感器的基线为 54cm，工作频率为 10Hz，校正后分辨率为 1240 376 像素。序列 00、02、05、06、07 和 09 包含循环。我们的 ORB-SLAM2 检测所有循环，并且能够在之后重用其地图，但序列 09 除外，其中循环发生在序列末尾的极少数帧中。表 I 显示了 11 个训练序列的结果，与最先进的立体 LSD-SLAM [11] 相比，这些序列具有公共地面实况，据我们所知，这是唯一显示所有序列详细结果的立体 SLAM。我们使用两种不同的度量，[3] 中提出的绝对平移 RMSE t_{abs} ，以及 [2] 中提出的平均相对平移 t_{rel} 和旋转 r_{rel} 误差。我们的

表 I KITTI 数据集中的准确性

比较。

Sequence	ORB-SLAM2 (stereo)			Stereo LSD-SLAM		
	t_{rel} (%)	r_{rel} (deg/100m)	t_{abs} (m)	t_{rel} (%)	r_{rel} (deg/100m)	t_{abs} (m)
00	0.70	0.25	1.3	0.63	0.26	1.0
01	1.39	0.21	10.4	2.36	0.36	9.0
02	0.76	0.23	5.7	0.79	0.23	2.6
03	0.71	0.18	0.6	1.01	0.28	1.2
04	0.48	0.13	0.2	0.38	0.31	0.2
05	0.40	0.16	0.8	0.64	0.18	1.5
06	0.51	0.15	0.8	0.71	0.18	1.3
07	0.50	0.28	0.5	0.56	0.29	0.5
08	1.05	0.32	3.6	1.11	0.31	3.9
09	0.87	0.27	3.2	1.14	0.25	5.6
10	0.60	0.27	1.0	0.72	0.33	1.5

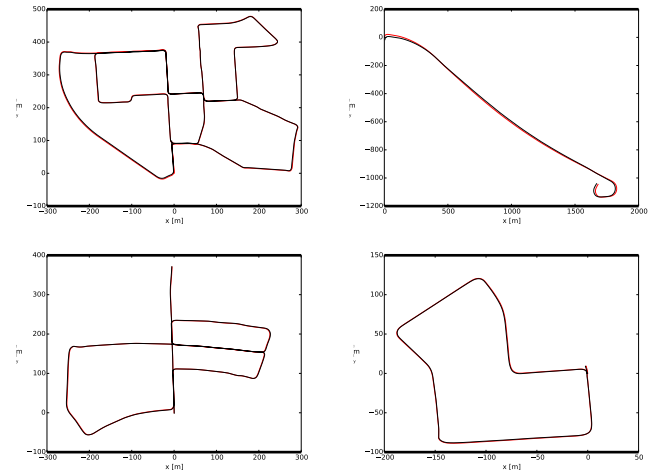


图 4. KITTI 00、01、05 和 07 中的估计轨迹（黑色）和地面实况（红色）。

系统在大多数序列中都优于 Stereo LSD-SLAM，并且总体上实现了低于 1% 的相对误差。序列 01，见图 3，是训练集中唯一的高速公路序列，平移误差稍差。在这个序列中平移很难估计，因为由于高速和低帧速率，很少有接近的点可以被跟踪。然而，由于有许多远点可以进行长距离跟踪，因此可以准确估计方位，实现每 100 米 0.21 度的误差。图 4 显示了估计轨迹的一些示例。

与 [1] 中提出的单目结果相比，所提出的立体版本能够处理单目系统失败的序列 01。在这个高速公路序列中，参见图 3，仅在少数帧中可以看到近距离点。立体版本能够仅从一个立体关键帧创建点，而不是延迟初始化单眼，包括查找两个关键帧之间的匹配，这一点对于该序列中不丢失跟踪至关重要。此外，立体系统以公制比例估计地图和轨迹，并且不会受到比例漂移的影响，如图 5 所示。

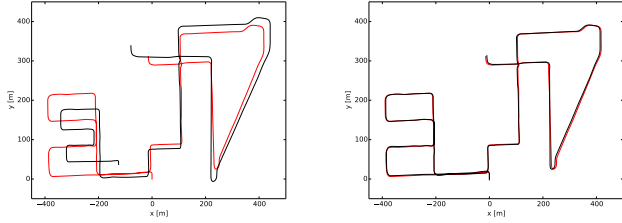


Fig. 5. Estimated trajectory (black) and ground-truth (red) in KITTI 08. Left: monocular ORB-SLAM [1], right: ORB-SLAM2 (stereo). Monocular ORB-SLAM suffers from severe scale drift in this sequence, especially at the turns. In contrast the proposed stereo version is able to estimate the true scale of the trajectory and map without scale drift.

B. EuRoC Dataset

The recent EuRoC dataset [21] contains 11 stereo sequences recorded from a micro aerial vehicle (MAV) flying around two different rooms and a large industrial environment. The stereo sensor has a $\sim 11\text{cm}$ baseline and provides WVGA images at 20Hz. The sequences are classified as *easy*, *medium* and *difficult* depending on MAV's speed, illumination and scene texture. In all sequences the MAV revisits the environment and ORB-SLAM2 is able to reuse its map, closing loops when necessary. Table II shows absolute translation RMSE of ORB-SLAM2 for all sequences, comparing to Stereo LSD-SLAM, for the results provided in [11]. ORB-SLAM2 achieves a localization precision of a few centimeters and is more accurate than Stereo LSD-SLAM. Our tracking get lost in some parts of *V2_03_difficult* due to severe motion blur. As shown in [22], this sequence can be processed using IMU information. Fig. 6 shows examples of computed trajectories compared to the ground-truth.

C. TUM RGB-D Dataset

The TUM RGB-D dataset [3] contains indoors sequences from RGB-D sensors grouped in several categories to evaluate object reconstruction and SLAM/odometry methods under different texture, illumination and structure conditions. We show results in a subset of sequences where most RGB-D methods are usually evaluated. In Table III we compare our accuracy to the following state-of-the-art methods: ElasticFusion [15], Kintinous [12], DVO-SLAM [14] and RGB-D SLAM [13]. Our method is the only one based on bundle adjustment and outperforms the other approaches in most sequences. As we already noticed for RGB-D SLAM results in [1], depthmaps for *freiburg2* sequences have a 4% scale bias, probably coming from miscalibration, that we have compensated in our runs and could partly explain our significantly better results. Fig. 7 shows the point clouds that result from backprojecting the sensor depth maps from the computed keyframe poses in four sequences. The good definition and the straight contours of desks and posters prove the high accuracy localization of our approach.

D. Timing Results

In order to complete the evaluation of the proposed system, we present in Table IV timing results in three sequences with

TABLE II
EuRoC DATASET. COMPARISON OF TRANSLATION RMSE (m).

Sequence	ORB-SLAM2 (stereo)	Stereo LSD-SLAM
V1_01_easy	0.035	0.066
V1_02_medium	0.020	0.074
V1_03_difficult	0.048	0.089
V2_01_easy	0.037	-
V2_02_medium	0.035	-
V2_03_difficult	X	-
MH_01_easy	0.035	-
MH_02_easy	0.018	-
MH_03_medium	0.028	-
MH_04_difficult	0.119	-
MH_05_difficult	0.060	-

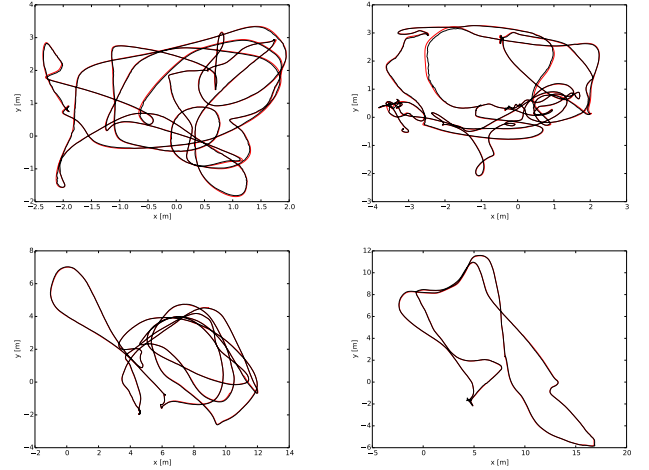


Fig. 6. Estimated trajectory (black) and groundtruth (red) in EuRoC V1_02_medium, V2_02_medium, MH_03_medium and MH_05_difficult.

different image resolutions and sensors. The mean and two standard deviation ranges are shown for each thread task. As these sequences contain one single loop, the full BA and some tasks of the loop closing thread are executed just once and only a single time measurement is reported. The average tracking time per frame is below the inverse of the camera frame-rate for each sequence, meaning that our system is able to work in real-time. As ORB extraction in stereo images is parallelized, it can be seen that extracting 1000 ORB features in the stereo WVGA images of V2_02 is similar to extracting the same amount of features in the single VGA image channel of fr3_office.

The number of keyframes in the loop is shown as reference for the times related to loop closing. While the loop in KITTI 07 contains more keyframes, the covisibility graph built for the indoor fr3_office is denser and therefore the loop fusion, pose-graph optimization and full BA tasks are more expensive. The higher density of the covisibility graph makes the local map contain more keyframes and points and therefore local map tracking and local BA are also more expensive.

V. CONCLUSION

We have presented a full SLAM system for monocular, stereo and RGB-D sensors, able to perform relocalization, loop

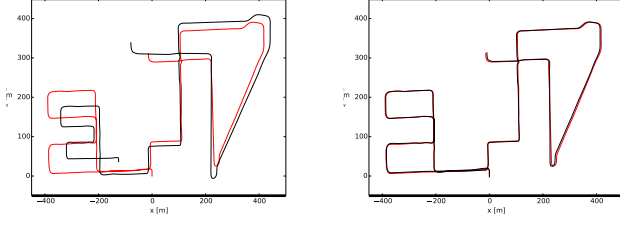


图 5. KITTI 08 中的估计轨迹 (黑色) 和地面实况 (红色)。左: 单目 ORB-SLAM [1], 右: ORB-SLAM2 (立体)。单目 ORB-SLAM 在这个序列中会遭受严重的尺度漂移, 尤其是在转弯处。相比之下, 所提出的立体版本能够估计轨迹和地图的真实比例, 而没有比例漂移。

B. EuRoC Dataset

最近的 EuRoC 数据集 [21] 包含 11 个立体序列, 这些序列是由微型飞行器 (MAV) 在两个不同的房间和一个大型工业环境中飞行时记录的。立体传感器的基线为 $\sim 11\text{cm}$, 并提供 20Hz 的 WVGA 图像。根据 MAV 的速度、照明和场景纹理, 序列被分类为 *easy*、*medium* 和 *difficult*。在所有序列中, MAV 都会重新访问环境, ORB-SLAM2 能够重用其地图, 并在必要时关闭循环。表 II 显示了与 Stereo LSD-SLAM 相比, ORB-SLAM2 对于所有序列的绝对平移 RMSE, 其结果在 [11] 中提供。ORB-SLAM2 的定位精度达到几厘米, 比 Stereo LSD-SLAM 更准确。由于严重的运动模糊, 我们的跟踪在 V2_03 *difficult* 的某些部分丢失。如 [22] 所示, 可以使用 IMU 信息来处理该序列。图 6 显示了计算轨迹与地面实况对比的示例。

C. TUM RGB-D Dataset

TUM RGB-D 数据集 [3] 包含来自 RGB-D 传感器的室内序列, 分为几个类别, 用于评估不同纹理、照明和结构条件下的对象重建和 SLAM/里程计方法。我们展示了通常评估大多数 RGB-D 方法的序列子集的结果。在表 III 中, 我们将我们的准确性与以下最先进的方法进行了比较: ElasticFusion [15]、Kintinously [12]、DVO-SLAM [14] 和 RGB-D SLAM [13]。我们的方法是唯一一种基于捆绑调整的方法, 并且在大多数序列中优于其他方法。正如我们在 [1] 中注意到的 RGB-D SLAM 结果, *freiburg2* 序列的深度图有 4% 的比例偏差, 可能来自校准错误, 我们在运行中对此进行了补偿, 并且可以部分解释我们明显更好的结果。图 7 显示了通过从四个序列中计算的关键帧姿势反投影传感器深度图而产生的点云。桌子和海报的良好清晰度和笔直轮廓证明了我们方法的高精度定位。

D. Timing Results

为了完成对所提出系统的评估, 我们在表 IV 中列出了三个序列的时序结果

表 II EUROC 数据集。翻译 RMSE

的比较 (m)。

Sequence	ORB-SLAM2 (stereo)	Stereo LSD-SLAM
V1_01_easy	0.035	0.066
V1_02_medium	0.020	0.074
V1_03_difficult	0.048	0.089
V2_01_easy	0.037	-
V2_02_medium	0.035	-
V2_03_difficult	X	-
MH_01_easy	0.035	-
MH_02_easy	0.018	-
MH_03_medium	0.028	-
MH_04_difficult	0.119	-
MH_05_difficult	0.060	-

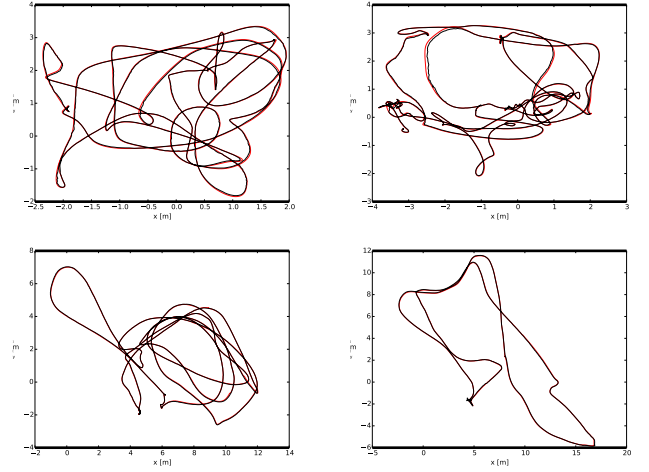


图 6. EuRoC V1 02 介质、V2 02 介质、MH 03 介质和 MH 05 困难中的估计轨迹 (黑色) 和真实值 (红色)。

不同的图像分辨率和传感器。显示每个线程任务的平均值和两个标准差范围。由于这些序列包含一个循环, 因此完整的 BA 和循环关闭线程的一些任务仅执行一次, 并且仅报告单个时间测量。每帧的平均跟踪时间低于每个序列的相机帧速率的倒数, 这意味着我们的系统能够实时工作。由于立体图像中的 ORB 提取是并行化的, 可以看出, 在 V2 02 的立体 WVGA 图像中提取 1000 个 ORB 特征与在 fr3 Office 的单个 VGA 图像通道中提取相同数量的特征类似。

显示循环中关键帧的数量, 作为与循环关闭相关的时间的参考。虽然 KITTI 07 中的循环包含更多关键帧, 但为室内 fr3 办公室构建的共视图更密集, 因此循环融合、位姿图优化和完整 BA 任务的成本更高。共视图的密度越高, 使得局部地图包含更多的关键帧和点, 因此局部地图跟踪和局部 BA 也更昂贵。

五、结论

我们推出了适用于单目、立体和 RGB-D 传感器的完整 SLAM 系统, 能够执行重定位、循环

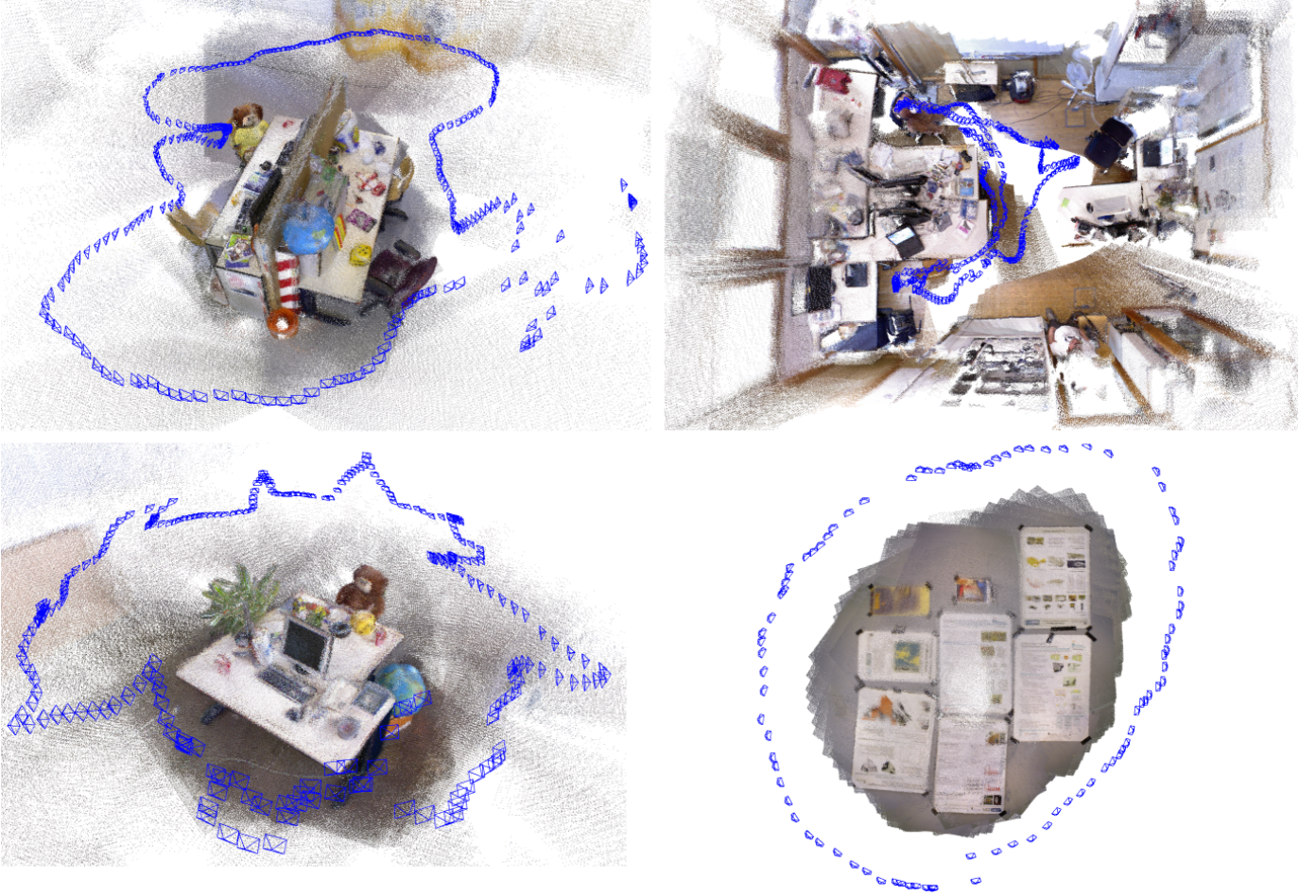


Fig. 7. Dense pointcloud reconstructions from estimated keyframe poses and sensor depth maps in TUM RGB-D *fr3_office*, *fr1_room*, *fr2_desk* and *fr3_nst*.

TABLE III
TUM RGB-D DATASET. COMPARISON OF TRANSLATION RMSE (m).

Sequence	ORB-SLAM2 (RGB-D)	Elastic- Fusion	Kintinuous	DVO SLAM	RGBD SLAM
fr1/desk	0.016	0.020	0.037	0.021	0.026
fr1/desk2	0.022	0.048	0.071	0.046	-
fr1/room	0.047	0.068	0.075	0.043	0.087
fr2/desk	0.009	0.071	0.034	0.017	0.057
fr2/xyz	0.004	0.011	0.029	0.018	-
fr3/office	0.010	0.017	0.030	0.035	-
fr3/nst	0.019	0.016	0.031	0.018	-

closing and reuse its map in real-time on standard CPUs. We focus on building globally consistent maps for reliable and long-term localization in a wide range of environments as demonstrated in the experiments. The proposed localization mode with the relocalization capability of the system yields a very robust, zero-drift, and lightweight localization method for known environments. This mode can be useful for certain applications, such as tracking the user viewpoint in virtual reality in a well-mapped space.

The comparison to the state-of-the-art shows that ORB-SLAM2 achieves in most cases the highest accuracy. In the

KITTI visual odometry benchmark ORB-SLAM2 is currently the best stereo SLAM solution. Crucially, compared with the stereo visual odometry methods that have flourished in recent years, ORB-SLAM2 achieves zero-drift localization in already mapped areas.

Surprisingly our RGB-D results demonstrate that if the most accurate camera localization is desired, bundle adjustment performs better than direct methods or ICP, with the additional advantage of being less computationally expensive, not requiring GPU processing to operate in real-time.

We have released the source code of our system, with examples and instructions so that it can be easily used by other researchers. ORB-SLAM2 is to the best of our knowledge the first open-source visual SLAM system that can work either with monocular, stereo and RGB-D inputs. Moreover our source code contains an example of an augmented reality application² using a monocular camera to show the potential of our solution.

Future extensions might include, to name some examples, non-overlapping multi-camera, fisheye or omnidirectional cameras support, large scale dense fusion, cooperative mapping or increased motion blur robustness.

²<https://youtu.be/kPwy8yA4CKM>

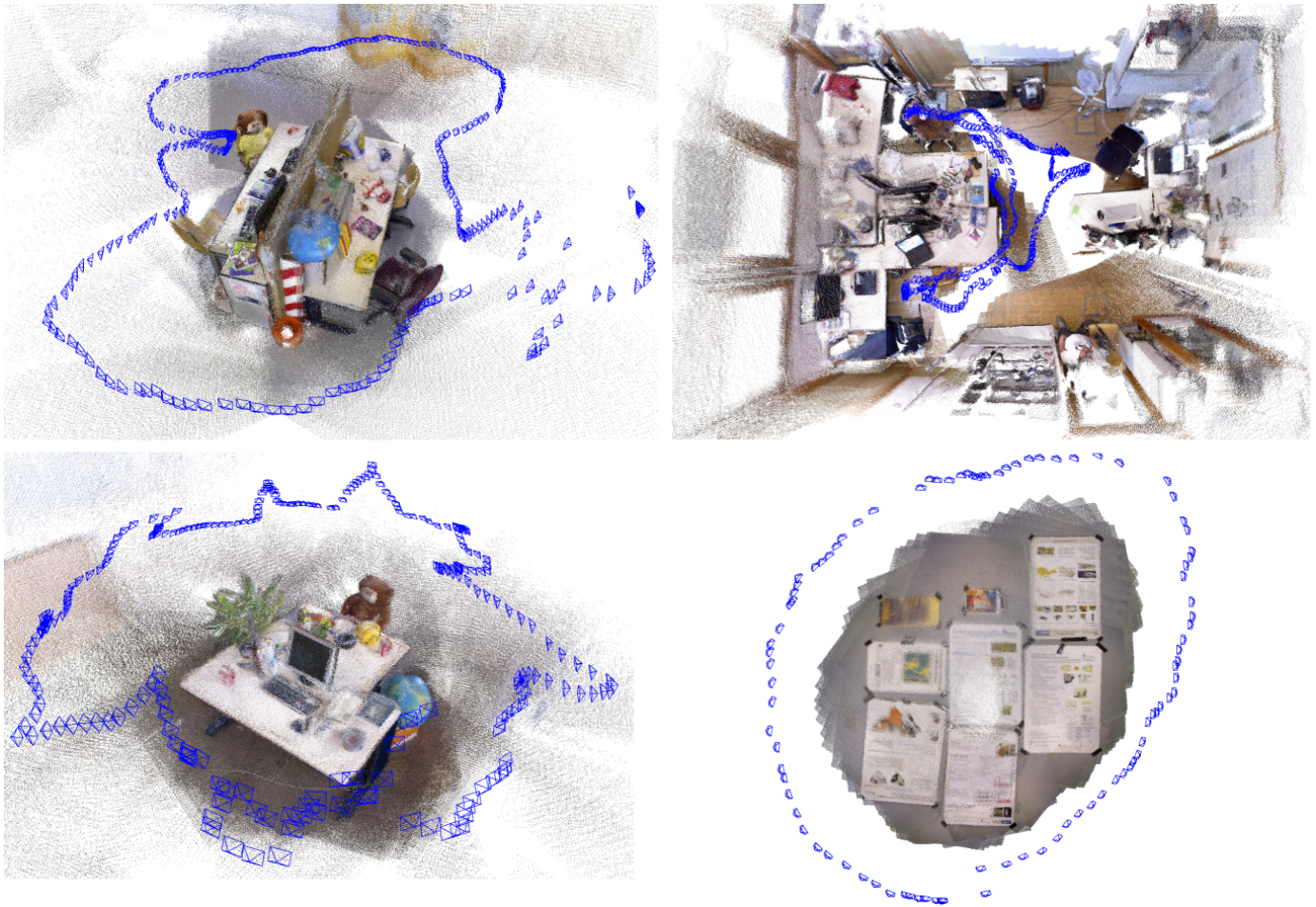


图 7. 密集点云侦察

来自 TUM R 中估计关键帧姿势和传感器深度图的指令

GB-D *fr3_office*、*fr1_room*、*fr2_desk* 和 *fr3_nst*

根误差 (m) 的比较。

表 III TUM RGB-D 数据集。翻译均方

Sequence	ORB-SLAM2 (RGB-D)	Elastic-Fusion	Kintinuous	DVO SLAM	RGBD SLAM
fr1/desk	0.016	0.020	0.037	0.021	0.026
fr1/desk2	0.022	0.048	0.071	0.046	-
fr1/room	0.047	0.068	0.075	0.043	0.087
fr2/desk	0.009	0.071	0.034	0.017	0.057
fr2/xyz	0.004	0.011	0.029	0.018	-
fr3/office	0.010	0.017	0.030	0.035	-
fr3/nst	0.019	0.016	0.031	0.018	-

在标准 CPU 上实时关闭并重用其地图。正如实验所证明的那样，我们专注于构建全球一致的地图，以便在各种环境中进行可靠和长期的定位。所提出的具有系统重定位能力的定位模式为已知环境提供了一种非常鲁棒、零漂移和轻量级的定位方法。此模式对于某些应用程序非常有用，例如在映射良好的空间中跟踪虚拟现实中的用户视点。

与最先进技术的比较表明，ORB-SLAM2 在大多数情况下实现了最高的精度。在

KITTI视觉里程计基准ORB-SLAM2是目前最好的立体SLAM解决方案。至关重要的是，与近年来蓬勃发展的立体视觉里程计方法相比，ORB-SLAM2 在已绘制地图的区域中实现了零漂移定位。

令人惊讶的是，我们的 RGB-D 结果表明，如果需要最准确的相机定位，捆绑调整的效果比直接方法或 ICP 更好，并且具有计算成本较低、不需要 GPU 处理实时运行的额外优势。

我们已经发布了我们系统的源代码，并附有示例和说明，以便其他研究人员可以轻松使用。据我们所知，ORB-SLAM2 是第一个开源视觉 SLAM 系统，可以使用单目、立体和 RGB-D 输入。此外，我们的源代码包含一个使用单目相机的增强现实应用程序²的示例，以展示我们解决方案的潜力。

举一些例子，未来的扩展可能包括非重叠多摄像头、鱼眼或全向摄像头支持、大规模密集融合、协作映射或增强的运动模糊鲁棒性。

²<https://youtu.be/kPwy8yA4CKM>

TABLE IV
TIMING RESULTS OF EACH THREAD IN MILLISECONDS (MEAN $\pm$ 2 STD. DEVIATIONS).

Settings	Sequence	V2_02	07	fr3_office
	Dataset	EuRoC	KITTI	TUM
	Sensor	Stereo	Stereo	RGB-D
	Resolution	752 $\times$ 480	1226 $\times$ 370	640 $\times$ 480
	Camera FPS	20Hz	10Hz	30Hz
	ORB Features	1000	2000	1000
Tracking	Stereo Rectification	3.43 $\pm$ 1.10	-	-
	ORB Extraction	13.54 $\pm$ 4.60	24.83 $\pm$ 8.28	11.48 $\pm$ 1.84
	Stereo Matching	11.26 $\pm$ 6.64	15.51 $\pm$ 4.12	0.02 $\pm$ 0.00
	Pose Prediction	2.07 $\pm$ 1.58	2.36 $\pm$ 1.84	2.65 $\pm$ 1.28
	Local Map Tracking	10.13 $\pm$ 11.40	5.38 $\pm$ 3.52	9.78 $\pm$ 6.42
	New Keyframe Decision	1.40 $\pm$ 1.14	1.91 $\pm$ 1.06	1.58 $\pm$ 0.92
	Total	41.66 $\pm$ 18.90	49.47 $\pm$ 12.10	25.58 $\pm$ 9.76
Mapping	Keyframe Insertion	10.30 $\pm$ 7.50	11.61 $\pm$ 3.28	11.36 $\pm$ 5.04
	Map Point Culling	0.28 $\pm$ 0.20	0.45 $\pm$ 0.38	0.25 $\pm$ 0.10
	Map Point Creation	40.43 $\pm$ 36.10	47.69 $\pm$ 29.52	53.99 $\pm$ 23.62
	Local BA	137.99 $\pm$ 248.18	69.29 $\pm$ 61.88	196.67 $\pm$ 213.42
	Keyframe Culling	3.80 $\pm$ 8.20	0.99 $\pm$ 0.92	6.69 $\pm$ 8.24
	Total	174.10 $\pm$ 278.80	129.52 $\pm$ 88.52	267.33 $\pm$ 245.10
Loop	Database Query	3.57 $\pm$ 5.86	4.13 $\pm$ 3.54	2.63 $\pm$ 2.26
	SE3 Estimation	0.69 $\pm$ 1.82	1.02 $\pm$ 3.68	0.66 $\pm$ 1.68
	Loop Fusion	21.84	82.70	298.45
	Essential Graph Opt.	73.15	178.31	281.99
	Total	108.59	284.88	598.70
BA	Full BA	349.25	1144.06	1640.96
	Map Update	3.13	11.82	5.62
	Total	396.02	1205.78	1793.02
Loop size (#keyframes)		82	248	225

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表 IV 每个线程的时序结果（以毫秒为单位）
（平均值 $\pm$ 2 标准偏差）。

S e n s i n g	Sequence	V2_02	07	fr3_office
	Dataset	EuRoC	KITTI	TUM
	Sensor	Stereo	Stereo	RGB-D
	Resolution	752 $\times$ 480	1226 $\times$ 370	640 $\times$ 480
	Camera FPS	20Hz	10Hz	30Hz
	ORB Features	1000	2000	1000
R e c t i f i c a t i o n	Stereo Rectification	3.43 $\pm$ 1.10	-	-
	ORB Extraction	13.54 $\pm$ 4.60	24.83 $\pm$ 8.28	11.48 $\pm$ 1.84
	Stereo Matching	11.26 $\pm$ 6.64	15.51 $\pm$ 4.12	0.02 $\pm$ 0.00
	Pose Prediction	2.07 $\pm$ 1.58	2.36 $\pm$ 1.84	2.65 $\pm$ 1.28
	Local Map Tracking	10.13 $\pm$ 11.40	5.38 $\pm$ 3.52	9.78 $\pm$ 6.42
	New Keyframe Decision	1.40 $\pm$ 1.14	1.91 $\pm$ 1.06	1.58 $\pm$ 0.92
	Total	41.66 $\pm$ 18.90	49.47 $\pm$ 12.10	25.58 $\pm$ 9.76
M a p p o i n t	Keyframe Insertion	10.30 $\pm$ 7.50	11.61 $\pm$ 3.28	11.36 $\pm$ 5.04
	Map Point Culling	0.28 $\pm$ 0.20	0.45 $\pm$ 0.38	0.25 $\pm$ 0.10
	Map Point Creation	40.43 $\pm$ 36.10	47.69 $\pm$ 29.52	53.99 $\pm$ 23.62
	Local BA	137.99 $\pm$ 248.18	69.29 $\pm$ 61.88	196.67 $\pm$ 213.42
	Keyframe Culling	3.80 $\pm$ 8.20	0.99 $\pm$ 0.92	6.69 $\pm$ 8.24
	Total	174.10 $\pm$ 278.80	129.52 $\pm$ 88.52	267.33 $\pm$ 245.10
P o s e	Database Query	3.57 $\pm$ 5.86	4.13 $\pm$ 3.54	2.63 $\pm$ 2.26
	SE3 Estimation	0.69 $\pm$ 1.82	1.02 $\pm$ 3.68	0.66 $\pm$ 1.68
	Loop Fusion	21.84	82.70	298.45
	Essential Graph Opt.	73.15	178.31	281.99
	Total	108.59	284.88	598.70
A B	Full BA	349.25	1144.06	1640.96
	Map Update	3.13	11.82	5.62
	Total	396.02	1205.78	1793.02
Loop size (#keyframes)		82	248	225

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